Final Exam of First Term (2023-2024)

Course name: Discrete Mathematical Structures	Academic year: 2023/2024
Course code: BAS61101	Date: 9/1/2024
Year and Section: First year- Computer Sciences	Time: 2 Hrs.
Course Instructor: Assist. Prof. Saeed Hemeda	Full Mark: 70. No of Questions: 3 No of Pages: 3

Answer the following Questions:

Question [1]: Choose the correct answer: **(20 marks)**

1) If A={1,2,3,4} and R is a relation on A such that: $R = \{(1,2),(1,3),(1,4)\}$, then the converse of R is R^T=.....

a) $\{(2,2),(1,1),(3,3),(4,4)\}$
b) $\{(2,1),(3,1),(4,1)\}$
c) $\{(2,1),(1,1),(3,1),(4,1)\}$
d) ϕ

❖ If A={0,1,2}, R and S are two relations on A where; $R = \{(0,2),(1,0),(1,2),(2,0)\}$ and $S = \{(1,0),(1,1),(1,2),(2,2)\}$, then;

2) M_R=.....

a) $\begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 2 & 1 & 0 & 0 \end{array}$
b) $\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$
c) $\begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$
d) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

3) M_S=.....

a) $\begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 2 & 1 & 0 & 0 \end{array}$
b) $\begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 2 & 0 & 0 & 1 \end{array}$
c) $\begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$
d) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

4) The statement: $[\sim(p \wedge \sim q)]$ is equivalent to .....

a) $p \rightarrow q$ b) $(p \wedge \sim q)$ c) $(p \vee q)$ d) $\emptyset$

5) Let $S = \{v_0, v_1, v_2, \ldots, v_n\}$ be the set of variables in a certain computer program. If $v_i = v_j$ means the values of v_i and v_j are the same, then at any moment during the course of running the program "=" is ..... relation on S.

a) reflexive b) symmetric c) transitive d) an equivalence

6) The statement $(p \to q) \vee (p \to \sim q)$ is .....

a) T b) F c) T and F d) $\sim p$

7) If $A = \{1,2,3\}, B = \{a,b,c,d\}; f = \{(1,a),(2,a),(3,c)\}$ then f is .....

a) onto b) injection c) bijection d) not injection

8) The statement $p \to (p \vee q)$ is .....

a) T b) F c) T and F d) T or F

❖ If R is a relation on $A = \{1,2,3,4\}$ and $R = \{(1,2),(1,1),(2,4),(3,1),(4,3)\}$;

9) R^T=.....

a) $\{(2,1),(1,1),(4,2),(3,3),(3,1)\}$
b) $\{(1,3),(1,4),(2,1),(2,2),(3,2)\}$
c) $\{(2,1),(3,2),(4,2),(2,5),(3,4)\}$
d) $\{(2,1),(1,1),(4,2),(1,3),(3,4)\}$

10) M_{R^c}=.....

a) $\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$

d) $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

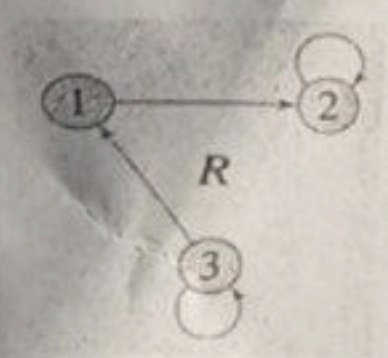
M_R $\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 1 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 & 1 \\ 3 & 1 & 0 & 0 & 0 \\ 4 & 0 & 0 & 1 & 0 \end{array}$

Question [2]:

(20 marks)

1) Using the laws of Boolean algebra to prove that:
$(p+q)(p+r) = p + qr$;
where p, q, r are three logical statements.

2) If $A = \{1,2,3\}$ and R is a relation on A defined by the following diagram:



Find the relation $(\overline{R^T} - R)$ and its matrix and graph its diagram.

Question [3]: **(30 marks)**

1) If $A = \{3,2,4,1\}$, $B = \{b,a,c\}$, $C = \{x,y\}$ and

$R_1 = \{(1,a),(2,a),(1,b),(2,b),(2,c),(3,b),(4,a)\}$,

$R_2 = \{(1,a),(2,a),(1,b),(2,b),(2,c),(3,b),(4,a)\}$, find $M_{(R_1 + R_2)}$.

2) If $A = \{5,3,2,4,1\}$, $R = \{(1,2),(2,3),(2,4),(2,5),(4,3),(4,5),(5,1),(5,4)\}$, draw the diagram and write the cycle of length 4.

3) Let $f: R \to R; f(x) = e^x$, Discuss the type of function (Onto-injection-bijection).

With My Best Wishes